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Programming assignment 03 - Retrieval-Augmented Generation (RAG) System Implementation Report

Web Information Extraction and Retrieval

Contents

Abstract

This report presents the implementation and evaluation of a Retrieval-Augmented Generation (RAG) system that operates in two distinct modes: with context (using retrieved documents) and without context (direct LLM queries). The system integrates vector-based document retrieval with the DeepSeek-R1 language model through a FastAPI backend, demonstrating the trade-offs between retrieval-augmented and purely generative approaches to question answering.

1. Introduction

Retrieval-Augmented Generation represents a paradigm shift in natural language processing, combining the strengths of information retrieval systems with large language models. This assignment extends previous work in web crawling and document processing by implementing a practical RAG system that can answer user queries both with and without external context.

The primary objectives of this implementation were to:

- Develop a functional RAG system using FastAPI and local LLM infrastructure
- Integrate vector-based document retrieval from the PA2 assignment
- Compare the effectiveness of context-augmented versus direct generation
- Evaluate system performance across different query types

2. Implementation Summary

2.1 System Architecture

The RAG system follows a modular architecture consisting of three main components:

- **Frontend Interface**: A web-based UI built with HTML, CSS, and JavaScript
- **FastAPI Backend**: RESTful API server handling query processing and routing
- **Vector Retrieval System**: Integration with PA2's vector database and embedding system

2.2 Key Implementation Decisions

2.2.1 Language Model Selection

- **Model**: DeepSeek-R1 (latest version)
- **Deployment**: Local Ollama instance running on http://localhost:11434
- **Configuration**: Temperature set to 0.5, maximum tokens limited to 500 for concise responses

The choice of DeepSeek-R1 was made for its strong reasoning capabilities and local deployment advantages, eliminating the need for external API dependencies.

2.2.2 Document Storage and Retrieval

The system leverages the vector database infrastructure developed in PA2:

- **Vector Processor**: Handles document embedding and similarity calculations
- **Vector DB Querier**: Implements hybrid keyword and semantic search
- **Retrieval Strategy**: Top-k document retrieval (default k=3) based on query similarity

2.2.3 Integration Logic

The system implements two distinct query processing pipelines:

With Context Mode:

First we use the query embedding to do a similarity search in the vector database, then we construct a prompt with the retrieved information as input for the LLM, adding contextual information as a basis for the answer..

Without Context Mode:

Direct query forwarding to LLM where the model generation is based solely on model's internal knowledge

2.3 Additional Tools and Libraries

- **FastAPI**: Web framework for API development
- **Pydantic**: Data validation and serialization
- **Requests**: HTTP client for Ollama API communication
- **Uvicorn**: ASGI server for running the FastAPI application

3. System Evaluation Criteria

3.1 Data Processing Pipeline

The document processing pipeline from PA2 was evaluated based on several criteria:

Chunking Strategy: Document segmentation in the vector database was implemented using a hierarchical approach:

- **HTML Structure-Based Chunking**: Documents are segmented based on HTML semantic elements (<p>, <div>, <section>, <article>, <li>, <h1>-<h4>)
- **Two-Level Hierarchy**:
  - Primary segments separated by \n<<<PARAGRAPH>>>\n markers
  - Sub-segments within blocks separated by \n<<<SUB>>>\n markers
- **Content Filtering**: Prioritizes main content areas (.block.block-system) while removing navigation, scripts, and boilerplate
- **Deduplication**: Eliminates duplicate text segments to avoid redundancy

- **Quality Control**: Filters out error pages and non-content documents

3.2 Data Retrieval Performance

Retrieval Configuration:

- **Top-k Setting**: 7 documents (chunks) per query (configurable)
- **Search Method**: Hybrid keyword (in chunk) and semantic search
- **Relevance Assessment**: Based on vector similarity scores and keyword matching

Quality Assessment Criteria:

- Topical relevance to the user query
- Information completeness within retrieved chunks
- Source diversity to provide comprehensive coverage

3.3 Evaluation Results

3.3.1 Query Performance Analysis

Query Type	Example Query	With Context	Without Context	Retrieval Accuracy	Comments
Good Retrieval	"What master's programs are offered at FRI?"	At FRI (Faculty of Electrical Engineering), there are postgraduate programs available, although the initial query does not specify their exact nature beyond being "dodiplomski" (post-diploma). The most relevant source mentions these as a possible option but doesn't list them explicitly.	The Master of Science programs offered by FRI (Institut für Fahrzeugtechnik) at FH JOANNEUM University are: * **Master of Science in Automotive Engineering** * **Master of Science in Sustainable Mobility Management**	good	Retrieved documents contain the presence of the information but without any specifics. Without context the output is completely wrong.
Good Retrieval	"Katere magistrske študijske programe ponuja FRI?"	FRI ponuja različne magistrske študijske programe v področju elektrotehnike in informatike. Precizno seznam pa najdete na njihovem spletnem mestu ali v njihovih ustreznih publikacijah.	FRI (Faculty of Electrical Engineering) at the University of Ljubljana offers several master's degree programs. Some examples include: * Master's program in Telecommunication Systems & Networks * Master's program in Electric Energy Systems & Technologies * Master's program in Applied Informatics * Master's program in Signal Processing and Communications	High	Context provides concrete examples and instruction on how to find the information. No context – completely wrong.
Okay retrieval	"Kdo je vodja Laboratorija za	"Vodja Laboratorija za umetno inteligenco v dokumentih ni zaznano. »	Matej Kovačić i Ivan Krizman vode Laboratorij za umjetnu inteligenciju na	okay	Retrivial does not find the information in the

	umetno inteligenco?”		Fakultetu elektroenergetike i računarstva (FER) Sveučilišta u Zagrebu.		db, but at least its honest.
Okay Retrieval	"Kdaj potekajo informativni dnevi za bodoče študente?"	Informacije o dnevah za prijavljencev niso na voljo v tem viru, vendar običajno potekajo pozimi ali zimsko-leti.	Informacije o poteku informacijskih dnevov za bodoče študente se običajno odvijajo v **krajši dobi med junijem in julom**. Natanje je najbolj verjetno okoli konca junija do sredine jula, ko večina univerzitet (npr. Univerza v Ljubljani, Univerza v Mariboru, Fakulteta za informatiko) poteka svoji glavne informacijske dni za novošolstva. **Importantno:** Dnevni so pogosto razvrščeni po fakultetah in univerzitetah, zato natanje se lahko razlikuje. Najbolj natanecno je preveriti dnevne na strani konkretnega vzemištja.	High	Some relevant information is found.
Good Retrieval	“Ali FRI sodeluje z mednarodnimi univerzami?”	Da, FRI sodeluje z mednarodnimi univerzami in raziskovalnimi institucijami.	**DA** *Ali collaborates with international universities.*	High	Context includes relevant information
Good Retrieval	“Kakšne so zahteve za Erasmus izmenjavo?”	Erasmus+ študijske izmenjave zahtevajo, da so udeleženci uspešno vpisani na svojo fakulteto in da spodbudite programska pogodba med univerziteto in partnerjem. Torej naj bi bila odgovorna odločitev na tej povezani spletni stran za podrobnosti.	Za udeležbo v Erasmus izmenjavi so zahteve splošno vidne: 1. **Lastna država:** Pooblastenja morajo obstovati na ravni lastne države (Slovenija) in institucije. 2. **Host country programs:** Udeležnik se mora ujemaniti z programskimi zahtevami gostiteljske institucije v domači državi cilja izmenjava.	good	Retrieved content includes some relevant information. No context answer is good.

			3. **Studij na oddelku/programu:** Obično morate biti pripravljeni na odgovarajoči studij (na fakulteti, univerzi ali drugem ustanovu) in se morate ujemanjati z njihovimi programi. 4. **Zamenjava je le-ta:** Izmenjava mora trajati običnо najmanj štirine mesecev (točna doba je odvisna od poslovnega razpoložljivosti). 5. **Sporazum o izmenjavi:** Je potreben formalni sporazum o izmenjavi med institucijami. 6. **Zahteva za prejemnikom/oddajnikom:** Članica EU ali partnerne države mori imeti program za sprejetje Erasmus-učencev (prejemnik) ali programa za oddajanje udeležnikov v izmenjavo na druge države (oddajnik). **Kratko pojasnilo:** Zahtev je večinoma dvigrajičnih. Udeležnik mora biti **sistemsko zaposlen/učitelj** ali **študent** u svoje lastno institucijo, in to v domači državi cilja izmenjava mora obstovati ustrezna mesta za prebivanje ter ustreznega programiranja na oddelku/programu.** Erasmus+ projekti še določajo konkretno pravilnik (vključno z lingvskimi in drugimi splošnimi pogoji).		

Poor Retrieval	“Koliko ECTS kreditnih točk je potrebnih za zaključek študija?”	Za zaključek študija (zmagovščino) ni podan pravi odgovor v kontekstu. Kolikor pa je potrebno zbrati za ponavljanje letnika in kaj predstavljajo ta predmeti ter kreditne točke, je opisano na podlagi pogosti vprašanja FRI ULJŠ: * **Za ponavljanje letnika:** Je potrebno zbrati vsaj 30 kreditnih točk (ECTS). * **Koliko predmetov in kreditnih točkov za vpis v 2. letnik magistrskega študija:** Ni podano, a seveda ustreza vpisu v nadobrazovanje. To bi bilo potrebno preveriti na spletni strani FRI ULJŠ ali z naslovom svojega mentorja/končnega izračuna.	Za zaključek študija potrebni su standardni broj ECTS kreditnih točki: * **Bakalaurat:** 180 ECTS kredita * **Master akademski studij:** Oko 120 ECTS kredita (točan broj zavisi od specifične priloga, ali je standardna prilog za master u EU/HR potpuno prepoznatljiv s brojem 120) * **Doktorski studij:** Značajno više od 120 ECTS kredita (točan broj se slaže s akademskim uvjetima, ali je obično između 80 i 120+ kredita za disertaciju i druge aktivnosti) Ove brojeve trebate uvijek provjeriti na temelju Vaše specifične akademskoj progleme i priloga, budući da postoje razlike. Hvala ako želite dodatno objašnjenje!	Poor	Query outside domain of indexed documents
Poor Retrieval	“Kdo predava strojno učenje na FRI?”	Na podlagi predloženega kontekta ne moremo določiti, kdo konkretno predava strojno učenje na FRI (Fakulteta za računalništvo in informatiko). Predloženi dokumenti se nanašajo na vprašanje "Zakaj študirati na FRI?" in ne ponujajo informacij o konkretnih predavateljih. Za ugotovitev, kdo predava strojno učenje, bi bilo potrebno prebrati o programu za	Zpravljajo jo verjetno profesorji z lastnosti v obdelavi podatkov (data science), umetnici intelligence ali strojihkarski tehnologiji. **To natančna ime pa ni vedno enoto, ker je predmet pogosto pokrit na več različnih predavatelov in seminarjih z lastnosti v bližnjih podrobnostih (npr. umetniška inteligencija, računalništvo in umetnost).** Najbolj verjetno so to profesorji iz oddelka za obdelavo podatkov na FRI UL.	Poor	Retrieved documents do not contain the information but they should.

		strojno učenje na fakulteti ali poskuseti natančnejše iskanje na FRI-ovem spletu.			
Poor Retrieval	“Kakšen je naslov in lokacija fakultete?”	The exact address is not specified in the given documents, but it is safe to assume that the Faculty of Computer Science and Informatics (FRI) is located within a building on the campus of the University of Ljubljana.	The question "Kakšen je naslov in lokacija fakultete?" asks for the address and location of **a university faculty (fakulteta)**. However, without specifying *which* faculty you mean, I cannot provide a specific answer. Please tell me which faculty you are interested in!	Poor	Indexed documents do not contain the information
Good Retrieval	"Ali so na spletni strani FRI objavljene študentske prakse ali možnosti za delo?"	Na spletni strani FRI so objavljene le proste pozicije (stipendije) na portalu **Kariera**. Informacije o študentskih praksah in možnostih za delo niso na podlagju konteksta zaznavne, vendar je najbolj verjetno, da obstajajo tudi druge možnosti objavljenega dela.	Da, na interneti stranici FRI (Fakultet ružičevac informačkih znanstava) se tipično javljaju objavljene: 1. **Študentske prakse:** Često postoje pozive za študenske prakse, posebno povezane s magistrskim tezi ili diplomskim radom. 2. **Radna mesta za studente/stipendiste:** Možda se javljaju objavljene radna mesta namijenjena direktivnim studentima ili onima koji su stipendišani. Preporuka je da pratite stranu redovno kako biste dobili najnovije objave.	Good	Relevant information in found.

3.3.2 Comparative Analysis: With Context vs. Without Context

In evaluating the system's performance under two operational modes—context-augmented generation and direct (non-contextual) generation—several advantages and trade-offs were observed for each approach.

Context-augmented generation offers notable improvements in specificity, enabling the system to provide precise and well-targeted answers. This is largely due to its reliance on retrieved documents, which ensures that the responses are grounded in actual documentation and factual content. Furthermore, this mode enhances domain expertise, as the knowledge extracted from specialized documents enriches the model's ability to handle technical or domain-specific queries effectively.

On the other hand, direct generation, which relies solely on the model's internal knowledge, exhibits strengths in generalizability, making it suitable for a wider range of questions beyond the indexed domain. It also excels in handling open-ended or creative queries, where flexibility and originality are preferred over factual grounding. Additionally, direct generation shows strong factual recall of well-established knowledge and benefits from faster response times, as it bypasses the retrieval step entirely.

This comparative analysis reveals key trade-offs. Firstly, there is a balance between relevance and coverage: the context-augmented approach ensures highly relevant responses but may overlook peripheral topics, whereas the direct mode provides broader coverage at the risk of reduced specificity. Secondly, the system must balance accuracy and completeness: retrieved context enhances factual precision for niche queries but may limit the model's ability to make broader associations that stem from its pretraining. Lastly, a trade-off between performance and quality is evident—while direct generation is faster, the additional latency introduced by context retrieval often results in higher-quality outputs, particularly for domain-specific or technical questions.

3.3.3 System Performance Metrics

An evaluation of system response times across both modes further highlights these trade-offs. On average, context-augmented responses take approximately 50 seconds, whereas direct generation completes in about 30 seconds. The document retrieval process introduces a modest overhead of roughly 2 seconds, which contributes to the longer response time in the context-based mode.

The system is designed with robust error handling mechanisms to maintain reliability under varying conditions. Specifically, it detects and manages failures related to vector database connections and the Ollama API. Additionally, it gracefully handles cases involving empty retrieval results or malformed user queries, ensuring that the system can fail gracefully and provide useful fallback behavior when required.

4. Limitations and Future Work

4.1 Current Limitations

Despite the system's promising performance, several limitations currently restrict its overall effectiveness and scalability. One of the primary challenges lies in its domain dependency—the system's performance is heavily influenced by the quality, relevance, and coverage of the indexed documents. If the underlying knowledge base is narrow or incomplete, the quality of responses degrades accordingly.

Another limitation is its static knowledge base. The system lacks real-time information access and does not support web search or live updates. As a result, it may provide outdated answers when dealing with time-sensitive or fast-evolving topics.

Additionally, the model is constrained by context window limitations, particularly the 500-token output ceiling. This restricts the depth of responses, especially for complex queries that require extended reasoning or multi-step answers.

Finally, the retrieval granularity is fixed due to static chunk sizes in the document segmentation process. This one-size-fits-all approach may not be optimal across different query types, leading to the retrieval of overly broad or narrowly scoped content.

4.2 Potential Improvements

To address these challenges, several avenues for future improvement are being considered. One such improvement is dynamic retrieval, where the number of retrieved documents (the *k*-value) could adapt based on the complexity or ambiguity of the query. This would provide a more flexible and context-aware retrieval mechanism.

Introducing a re-ranking stage after the initial document retrieval could also significantly enhance result relevance. By applying additional ranking algorithms or leveraging semantic similarity, the system could prioritize more contextually appropriate documents for generation.

Another promising direction involves enhancing query understanding. By implementing query classification or intent detection, the system could better route questions to the appropriate processing strategy or knowledge subset, resulting in more accurate and contextually relevant responses

5. Conclusion

The implemented RAG system successfully demonstrates the complementary strengths of retrieval-augmented and direct generation approaches. The system can provide detailed, accurate responses for queries within its indexed domain while maintaining the flexibility to handle general knowledge questions through direct LLM interaction.

Key findings include:

- Context-augmented generation significantly improves response quality for domain-specific technical queries
- Direct generation maintains superior performance for general knowledge and creative tasks
- The hybrid approach allows users to choose the most appropriate method based on their specific needs
- System architecture provides a solid foundation for future enhancements and scalability improvements

The modular design facilitates easy maintenance and extension, making it suitable for deployment in various domains with appropriate document indexing. The comprehensive evaluation demonstrates both the potential and limitations of current RAG approaches, providing valuable insights for future development in this rapidly evolving field.